



Higher Diploma
in
Information Technology

Final Examination
Year 1, Semester 2 (2025)
July Intake

Emerging Technologies and Frameworks
(IT1204)

Duration: 2 Hours

Instructions to Candidates:

- ◆ This is a closed book examination.
- ◆ This paper contains 4 questions on 8 pages without the cover page.
- ◆ Answer all questions on the WORKBOOK provided.
- ◆ Read all questions before answering.
- ◆ The total marks obtainable for this examination is 100.

Question One

(25 marks)

Select the correct/most suitable answer for the given questions. Write the letter of the correct answer next to the question number in the workbook provided. (2.5×10 marks)

- a) Metcalfe's Law states that the value of a telecommunications network is proportional to the square of the number of connected users. If a network doubles its users, by what factor does its value increase according to Metcalfe's Law?
- A. 2 times
 - B. 4 times
 - C. 8 times
 - D. 16 times
 - E. 32 times
- b) In the context of 4D printing, which of the following best explains the fourth dimension?
- A. The ability of printed objects to change shape or function over time in response to external stimuli
 - B. Addition of augmented reality features to 3D objects
 - C. Use of higher-resolution printing technology
 - D. A method of copying objects
 - E. Integration of artificial intelligence into 3D-printed models
- c) What is the primary function of an actuator in an IoT system?
- A. To detect environmental changes and convert them into data
 - B. To store large amounts of sensor data
 - C. To convert energy into motion and apply force to something
 - D. To monitor data transmission in the cloud
 - E. To observe the environment
- d) Which quantum principle allows a qubit to exist in multiple states simultaneously?
- A. Superposition
 - B. Entanglement

- C. Interference
 - D. Decoherence
 - E. Multiposition
- e) Which of the following is an example of narrow (or weak) AI?
- A. A robot with human emotions and reasoning
 - B. An AI that can perform any intellectual task like a human
 - C. A voice assistant like Siri or Alexa
 - D. A general-purpose machine that learns anything without training
 - E. An AI capable of conscious thought and self-awareness
- f) Which industry was first significantly transformed during the Industrial Revolution?
- A. Automobile
 - B. Polythene
 - C. Chemical
 - D. Steel
 - E. Textile
- g) Which technology is NOT typically considered a key enabler of the Metaverse?
- A. Virtual Reality (VR)
 - B. Augmented Reality (AR)
 - C. Quantum Computing
 - D. Visual Reality
 - E. Blockchain
- h) Which of the following best defines the Internet of Things (IoT)?
- A. A virtual network used for social media interactions
 - B. A software solution to control electronic devices
 - C. A cloud based storage service for large data backups
 - D. A network of physical objects embedded with sensors, software, and connectivity to exchange data

- E. A collection of internet protocols for web development
- i) What does the term "immutability" mean in the context of blockchain?
- A. Data on the blockchain can be edited or deleted by authorized users
 - B. Transactions are processed instantly without delays
 - C. The blockchain operates without the need for internet connectivity
 - D. Once data is recorded on the blockchain, it cannot be altered or deleted
 - E. Blockchain data is automatically backed up to multiple cloud servers daily
- j) Which of the following best describes the primary goal of Artificial Intelligence (AI)?
- A. To create machines that can perform repetitive manual tasks faster than humans
 - B. To build hardware components that enhance computational speed
 - C. To design software that can store and retrieve large volumes of data efficiently
 - D. To replace all human jobs with automated robots regardless of task complexity
 - E. To develop systems that can simulate human intelligence, including learning, reasoning, and problem-solving

Question Two

(25 marks)

A developing country has launched a national initiative called “AgriFusion 2040,” aimed at revolutionizing its agricultural sector through advanced technologies such as cloud computing, 3D printing, artificial intelligence, blockchain, etc. As a digital transformation consultant, you have been invited to provide insights by answering the following questions at a ministerial briefing.

- a) Write 2 potential challenges of integrating cloud computing into rural agricultural settings. (4 marks)
- b) Explain how 3D printing could contribute to improving agricultural productivity. Give 2 examples. (5 marks)
- c) Write 3 advantages of using AI in agriculture compared to manual methods. (6 marks)
- d) State 2 ways blockchain could ensure trust and transparency in the food supply chain. (4 marks)
- e) The Internet of Things (IoT) is another key enabler in smart agriculture. (6 marks)
 - i. How can IoT devices be used to support precision farming in the AgriFusion 2040 initiative?
 - ii. Provide 2 specific examples of IoT applications in agriculture, and explain.

Question Three

(25 marks)

Please refer to the source code of the webpage given below. Answer the questions that follow.

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>ETF Final Exam - DOM Structure</title>
5  </head>
6  <body>
7
8    <h2>DOM Structure Representation</h2>
9
10   <ul>
11     <li>
12       Document
13       <ul>
14         <li>
15           Head
16           <ul>
17             <li>Title: ETF Final Exam 2025</li>
18           </ul>
19         </li>
20         <li>
21           Body
22           <ul>
23             <li>Heading: Final Exam</li>
24             <li>Paragraph: This is the ETF final exam paper.</li>
25           </ul>
26         </li>
27       </ul>
28     </li>
29   </ul>
30
31 </body>
32 </html>
```

Figure 1: HTML code of a webpage

- a) Draw the DOM structure of the webpage given in **Figure 1**. (6 marks)
- b) Refer to the provided webpage source code (Figure 1) and answer the following questions using your knowledge of DOM traversal. (4 marks)
- The root node of the DOM Tree.
 - Next Sibling of `<h2>` Tag.
 - What is the First child of the `<Ul>` element inside `<body>`?
 - What is the Last Child of the `<body>` element?
- c) Write the code segment to add another list item under the “Body” section of the DOM structure using jQuery. Demonstrate your knowledge of DOM tree traversal techniques in your solution. (3 marks)

- d) Consider the following XML structure in **Figure 2**. Convert this XML structure to an equivalent JSON format. (5 marks)

```
1  <?xml version="1.0" encoding="UTF-8"?>
2  <students>
3    <student>
4      <id>1001</id>
5      <name>Alice Perera</name>
6      <department>School of Computing</department>
7    </student>
8    <student>
9      <id>1002</id>
10     <name>Sunil Fernando</name>
11     <department>School of Computing</department>
12   </student>
13   <student>
14     <id>1003</id>
15     <name>Ruwan Silva</name>
16     <department>School of Computing</department>
17   </student>
18 </students>
```

Figure 2: XML structure representing student data

- e) Consider the above answer of JSON data, assume that a backend API endpoint is available at <https://example.com/api/students> on a Node.js server. You are required to send the above JSON data to this API using and AJAX request. Write the necessary jQuery AJAX code to send this data using a POST request. (5 marks)
- f) i. What is the purpose of using an AJAX POST request in a web application? (2 marks)
- ii. Explain briefly in the context of sending student details to a server.

Question Four

(25 marks)

- a) Define MongoDB. Explain its key advantages over traditional relational databases like SQL, focusing on schema design, scalability, and language integration. (5 marks)
- b) Assume you are developing a **patient management system** using MongoDB. (5 marks)
- i. List the four basic **CRUD** operations supported by MongoDB.
 - ii. Write MongoDB commands to do the following tasks.
 - a. Add new patient with name “Kamal”, age 25, and doctor “Dr. Nimal”.
 - b. Retrieve all patients, who are assigned to doctor “Dr. Nimal”.
 - c. Update Kamal’s age to 26.
 - d. Remove Kamal’s record from the collection.
- c) Describe the purpose of the following HTTP verbs and provide an example of how each is used in the context of a REST API. (5 marks)
- i. GET
 - ii. POST
 - iii. PUT
 - iv. PATCH
 - v. DELETE
- d) Using the tables provided below, clearly explain each pair of terms by using the listed features in each table. (10 marks)

- i. JavaScript vs jQuery

Features	JavaScript	jQuery
Definition		
Purpose		

ii. XML vs JSON

Features	XML	JSON
Definition		
Structure		
Readability		

iii. SQL vs NoSQL

Features	SQL	NoSQL
Structure		
Schema		
Examples		

iv. POST vs GET

Features	POST	GET
Purpose		
Visibility		
Use Case		

v. Bootstrap vs CSS

Features	Bootstrap	CSS
Definition		
Purpose		

End of the Question Paper